CEC 419: Design Project Proposal

Project Title: AI-Based Sign Language Recognition System

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Executive Summary:
The project aims to develop an AI-driven system that recognizes sign language gestures and translates them into text or speech in real-time. Using machine learning and computer vision (e.g., Convolutional Neural Networks), the system will process video or image input to facilitate communication between hearing-impaired individuals and those unfamiliar with sign language. The platform will be web-based, interactive, and user-friendly, promoting technological innovation and social inclusion in Cameroon.

Problem Statement:
Communication barriers limit deaf and speech-impaired individuals in accessing education, healthcare, and public services. Human interpreters are costly, scarce, and raise privacy concerns. The project proposes an AI solution for automatic real-time translation of sign language gestures.

Project Objectives:

- Develop a deep learning-based gesture recognition system.
- Translate gestures into text and speech in real-time.
- Enhance inclusivity and accessibility for the deaf community.
- Design an intuitive web or mobile interface.
- Adapt the system for Cameroonian and other African sign languages.

Tools and Technologies:

- Programming: Python
- ML Libraries: TensorFlow, Keras, Scikit-learn
- Computer Vision: OpenCV
- Data Handling/Visualization: NumPy, Pandas, Matplotlib
- Frontend: HTML, CSS, JavaScript, React.js
- Backend: Flask/FastAPI

- Database: SQLite/PostgreSQL
- Deployment: Render/Heroku/Localhost
- Version Control: Git & GitHub

Functional Applications:

- Education: Supports inclusive classrooms by translating sign language.
- Healthcare: Facilitates communication with deaf patients.
- Public Services: Improves interactions in offices, banks, and service points.
- Personal Communication: Enables daily communication for individuals.
- Training & Research: Supports AI and accessibility research.
- Accessibility: Adaptable for mobile and web use.

Proposed Supervisors: Mr. Baloko Collins and Mr. Tchagna